ADVANCED QUANTUM MECHANICS

Exercise 8 - 02.12.2025

TECHNISCHE UNIVERSITÄT DARMSTADT

1. Scattering from a 3D contact potential - Born approximation

Consider elastic scattering in three dimensions from a point-like interaction

$$V(\vec{r}) = \lambda\, \delta^{(3)}(\vec{r}),$$

with an incoming plane wave of momentum $\vec{k}_i = k\,\hat{z}$. Use the scattering amplitude in the first Born approximation

$$f^{(1)}(\theta,\phi) = -\frac{m}{2\pi\hbar^2}\int \mathrm{d}^3 r\, \mathrm{e}^{-\mathrm{i}(\vec{k}_f-\vec{k}_i)\cdot\vec{r}}\, V(\vec{r}), \qquad \vec{k}_f = k\,\hat{r}.$$

(i) to show that the scattering amplitude $f^{(1)}(\theta,\phi)$ is independent of the scattering angles.

(ii) Compute the differential and total cross sections,

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \left|f^{(1)}(\theta,\phi)\right|^2, \qquad \sigma^{(1)}_{\text{tot}} = \int \mathrm{d}\Omega\, \frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}.$$

2. Scattering from an impenetrable sphere

Consider elastic scattering from a hard sphere of radius a, defined by the potential

$$V(r) = \begin{cases} \infty, & r < a, \\ 0, & r > a. \end{cases}$$

The wavefunction must satisfy the boundary condition $\psi(r=a,\theta,\phi) = 0$. For $r > a$ the potential vanishes, and a partial–wave expansion is appropriate. The reduced radial function satisfies

$$\left[\frac{\mathrm{d}^2}{\mathrm{d}r^2} + k^2 - \frac{\ell(\ell+1)}{r^2}\right] u_\ell(r) = 0, \qquad r > a.$$

with general solution as a linear combination of spherical Bessel functions,

$$u_\ell(r) = A_\ell\, j_\ell(kr) + B_\ell\, n_\ell(kr).$$

(i) Impenetrability implies $u_\ell(a) = 0$. Prove that the phase shifts satisfy

$$\tan\delta_\ell(k) = \frac{j_\ell(ka)}{n_\ell(ka)}.$$

(ii) Argue that for $ka \ll 1$ only the s–wave contribution survives.

(iii) Using $j_0(x) = \sin x/x$ and $n_0(x) = -\cos x/x$, find $\delta_0(k)$

(iv) Insert this into the total cross section and prove that

$$\sigma_{\text{tot}} \xrightarrow{k\to 0} 4\pi a^2.$$

3. Born approximation for a finite spherical potential

Consider elastic scattering from the spherically symmetric potential

$$V(r) = \begin{cases} V_0, & r < a, \\ 0, & r > a. \end{cases}$$

Let $k = \sqrt{2mE}/\hbar$ and $q = |\vec{k}_f - \vec{k}_i| = 2k\sin(\theta/2)$.

(i) Starting from the Born formula for central potentials,

$$f^{(1)}(\theta) = -\frac{2m}{\hbar^2}\frac{1}{q}\int_0^\infty r\,V(r)\,\sin(qr)\,dr,$$

evaluate the integral for the square well and show that

$$f^{(1)}(\theta) = -\frac{2mV_0}{\hbar^2}\,\frac{\sin(qa) - qa\cos(qa)}{q^3}.$$

(ii) Write the angular distribution $\frac{d\sigma}{d\Omega}$ in the low–energy limit $qa \ll 1$. By expanding $\sin(qa)$ and $\cos(qa)$ to order $(qa)^3$, show that

$$f^{(1)}(\theta) \xrightarrow{qa\ll 1} -\frac{2mV_0}{\hbar^2}\,\frac{a^3}{3}, \qquad \text{(isotropic)}.$$

and the same for the differential cross section.

(iii) We will use the positions of the first three minima in the $^{40}\mathrm{Ca}$ data for proton scattering off this nucleus located at:

$$\theta_1 = 7.8^\circ, \qquad \theta_2 = 14.4^\circ, \qquad \theta_3 = 20.8^\circ,$$

for a proton kinetic energy of $E = 39\,\mathrm{MeV}$. Determine the radius of $^{40}\mathrm{Ca}$ from each of the angles $\theta_1, \theta_2, \theta_3$.

(iv) Compute the radius using the empirical formula

$$a_{\mathrm{emp}} = 1.4\,A^{1/3}\ \mathrm{fm} \qquad (A = 40),$$

and compare with the values obtained before.